

AKASH AGALAVE

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EDUCATION

Dr. D.Y. Patil University

B.Tech - Computer Science and Engineering | CGPA - 8.45

Dec 2021 - June 2025

Pune, India

Maharashtra Vidyalaya Barshi

Higher Secondary Education (Science) | Percentage - 86.85%

May 2019 - June 2021

Barshi, India

EXPERIENCE

Euron

August 2025

AI Developer Intern

Remote, India — Individual project-based

- Designed and implemented AI applications by developing models for a medical chatbot using domain-specific **data**.
- Implemented **vector search** for fast and relevant medical information retrieval.
- Developed backend services and application **systems**, containerized using Docker to support **scalable deployment**.
- Deployed on **AWS EC2** using images from AWS ECR with CI/CD automation.

CodersCave

May 2024 - June 2024

Data Science Intern

Remote, India — Individual project-based

- Developed and trained **machine learning models** on text data, achieving 80.7% accuracy.
- Streamlined text preprocessing, tokenization, and feature extraction for training pipeline.
- Evaluated and optimized **model performance**, delivering 0.72 F1-score.

PROJECTS

VeriRAG – Agentic Research Intelligence & Scientific Claim Verification | [GitHub](#)

May 2026

- Built an **agentic RAG** pipeline using LangGraph with **intelligent query routing** and hybrid BM25 + dense retrieval with RRF fusion, replacing static single-path retrieval workflows.
- Achieved **94% Faithfulness** and **88% Answer Relevancy** on DeepEval benchmarks, while implementing **prompt injection detection**, retrieval sanitization, and PII redaction to secure AI interactions.
- Built a Redis-based distributed cache for embeddings and document processing (**49% hit rate, 90%+ cost reduction**) and implemented circuit-breaker-backed **multi-provider LLM routing** (GPT-4o-mini → Claude Haiku → Gemini Flash) for high availability.
- Containerized with Docker, deployed on AWS EKS with HPA autoscaling, and automated **zero-downtime** releases through GitHub Actions CI/CD.

AutoML Brain – Automated ML Monitoring & Deployment Platform | [GitHub](#)

Jan 2026 - Apr 2026

- Built a system for monitoring model performance and system behavior by tracking latency, error rates, and simulated data drift (PSI), enabling automated decisions for scaling and retraining.
- Implemented a **multi-agent system** (Governance, Deployment) with a **7-stage DVC pipeline**, automating the ML lifecycle from data ingestion to monitoring and governance.
- Improved system performance by reducing **P95 latency by 82%** (2000ms → 350ms) and enabled safe model validation using **shadow-based canary** deployment with Prometheus-driven metrics.
- Deployed the system on AWS EKS using Docker, MLflow, and CI/CD pipelines, integrating **automated retraining** jobs, model registry, and production-grade deployment workflows..

FraudGuard – Real-Time Fraud Detection & MLOps Platform | [GitHub](#)

Nov 2025 – Jan 2026

- Mitigated **0.75% class imbalance** using LightGBM with cost-sensitive learning and F1-optimized threshold via precision-recall curve, achieving **PR-AUC 0.945**.
- Deployed FastAPI on EKS with async Kafka producers and custom Prometheus instrumentation, sustaining 1.2K RPS with model-only inference P95 24ms and P99 31ms..
- Architected a DVC pipeline with PSI drift detection that auto-dispatches GitHub Actions retraining with an MLflow quality gate before every Production promotion.
- Deployed containerized FastAPI microservices on AWS EKS with horizontal autoscaling and **zero-downtime rolling updates**.

TECHNICAL SKILLS

Languages: Python, SQL

Technologies: Machine Learning, Deep Learning, NLP, RAG, AI Agents

Frameworks & Libraries: PyTorch, Langchain, SHAP

Database: SQL, Vector Database(Qdrant)

MLOps-Tools: Git, GitHub, DVC, MLflow, Github Actions, DagsHub, Prometheus, Grafana

Deployment & Cloud Services: FastAPI, Docker, AWS, Kubernetes (EKS)